[PRINT]

MAT-225-16805-M01 Calc I: Single-Variable Calc 2025 C-6 (Oct - Dec) : MAT-225 : 2079955,
Aleksey Ahmann, 11/18/25 at 2:17:25 AM EST

Question1: *Score 1/1*

Introduction

The year is 3500 and your spaceship has crashed on the planet Fluxion. Fortunately, your computer has survived the impact, and you open it up to determine the next course of action. You learn that your computer has some maps and information on the civilization, and that the city (and perhaps the planet) was abandoned several years ago. Based on the maps, it appears that your only hope of reaching a communication device is to head to what appears to be the center of a city that you crashed near. Your computer warns you that the city is protected by several gates, which have passcodes to activate.

The maps and images show a strange alien language that you have never encountered. You ask your computer if it can translate this language and are told that the language, based on its structure, can only be translated into mathematics. The computer will translate the gates' riddles and any other information into mathematics. You will need to answer the riddle as posed by your computer (in mathematical language) for it to be able to enter the passcode for the gates. Fortunately, you have a notebook, pencil, and a mathematics text on your computer as supports. You take your supplies, take one last look at your broken ship, and head to towards the first gate.

The First Gate

Upon arriving at the first gate, you ask your computer to translate the riddle. "These appear to be limits", your computer replies. "If you provide me with the answer to each of below questions, I will compile, translate, and communicate the passcode to the gate. There is note here that the parameter a represents a real number such that $0 < a < 5$." Your computer also reminds you to type "infinity" for ∞, "-infinity" for $-\infty$, and "NA" if the limit does not exist. You grab your notebook and pencil and begin.

$$\lim_{x \to a^-} \frac{x+5}{x-a} =$$

Your response	Correct response
-infinity	**-infinity**

Auto graded **Grade:** 1/1.0

$$\lim_{x \to a^+} \frac{x+5}{x-a} =$$

Your response	Correct response
infinity	**infinity**

Auto graded **Grade:** 1/1.0

$$\lim_{x \to a} \frac{x+5}{x-a} =$$

Your response	Correct response
NA	**NA**

Auto graded **Grade:** 1/1.0

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

The problem, or "riddle," involves the evaluation of the limit, on both sides, of the expression:

$$f(x) = \left\{ 0 < a < 5 : \frac{x+5}{x-a} \right\}$$

In particular, whether-or-not the `f(x)`'s limit exists. Recall that a limit at x approaching an arbitrary value a is met with the following condition:

$$\lim_{x \to a} f(x) \equiv \left[\lim_{x \to a^-} \frac{x+5}{x-a} = \lim_{x \to a^+} \frac{x+5}{x-a} = L \right]$$

To demonstrate, or disprove, the limit of the given function, I have employed experimental techniques as opposed to the more rigourous solution. In particular, I have plotted the expression for different parameters for "a" within the given $0 < a < 5$ range.[1] I noticed that the function, when approached from the left, approaches negative infinity, and when approached from the right, approaches positive infinity. A vertical asymptote is formed towards the $x - a$ part of the fraction, though since the L-value of the $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$ the limit $\lim_{x \to a} f(x)$ simply does not exist.

Endnotes

1. Sadly, I cannot paste images into this writeup, so I will have to stick to a verbal explaination.

Ungraded **Grade:** 0/0.0

Total grade: 1.0×1/3 + 1.0×1/3 + 1.0×1/3 + 0.0×0/3 = 33% + 33% + 33% + 0%

Question2: *Score 1/1*

The Second Gate

As you move through the first gate, you can see another gate not far in front of you. You approach the second gate and your computer reads: "There are 2 values for which the below function does not exist. The passcode is the limit of $g(x)$ as x approaches the smaller of these two values."

$$g(x) = \frac{x+7}{x^2-49}$$

Your computer also reminds you to type "infinity" for ∞, "-infinity" for $-\infty$, and "NA" if the limit does not exist.

What do you enter for your computer to translate?

Your response	Correct response
$-\frac{1}{14}$	**-1/14**

Auto graded **Grade:** 1/1.0

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

This problem involves working out a "passcode" in the form of the solution to the limit of the following expression:

$$g(x) = \frac{(x+7)}{(x^2-49)}$$

In particular, the lesser constant from the factoring of this expression's divisor is the thing to evalaute the limit as x approaches the aforementioned lesser constant. The problem is therefore set up as:

$$\lim_{x \to \left[\operatorname{argmin}_{c_i \in \text{ factored}(x^2-49)}\right]} \frac{x+7}{x^2-49} = L$$

Such a solution is worked out as:

$$\text{Worked Solution} \longrightarrow \begin{cases} \lim_{x \to a} \frac{x+7}{x^2-49} & \because \text{ Given} \\ \lim_{x \to a} \frac{x+7}{(x+7)(x-7)} & \because x^2 - c = (x+c)(x-c) \\ \lim_{x \to a} \frac{\cancel{x+7}}{\cancel{(x+7)}\,(x-7)} & \\ \lim_{x \to -7} \frac{1}{(x-7)} & \because (x+7) \text{ cancels out,} \\ & \wedge \operatorname{argmin}_{c_i \in (-7,7)} = -7 \\ \lim_{x \to -7} \frac{1}{(-7-7)} = -\frac{1}{14} & \because \text{ Direct Substitution} \end{cases}$$

Thus, I have demonstrated that the **L**-value for the given limit of the given expression is:

$$\lim_{x \to a} f(x) = -\frac{1}{14}$$

Ungraded **Grade:** 0/0.0

Total grade: 1.0×1/1 + 0.0×0/1 = 100% + 0%

Question3: *Score 1/1*

The Third Gate

You arrive at the third gate and look for the inscription. Your computer translates the following:

"This passcode consists of the two below limits which include an arbitary real number, c. Remember to type "infinity" for ∞, "-infinity" for $-\infty$, and "NA" if the limit does not exist. Enter the results and I will apply the passcode."

$$\lim_{x \to c^+} \frac{22}{(x-c)^9} =$$

Your response	Correct response
infinity	**infinity**

Auto graded **Grade:** 1/1.0

$$\lim_{x \to c^-} \frac{22}{(x-c)^9} =$$

Your response	Correct response
-infinity	**-infinity**

Auto graded **Grade:** 1/1.0

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

This problem involves working out the limit for both left and right sides for the given expression:

$$f(x) = \frac{22}{(x-c)^9}$$

with **c** taking the form of any real number, or $\mathbb{D}_{f(x)} = \{-\infty < c < +\infty\}$

Which is set up as:

$$\lim_{x \to c^-} f(x) = L_1$$
$$\wedge \lim_{x \to c^-} f(x) = L_2$$

I did not give a formal solution to this problem. Instead, I employed the graphical method,[1] which I then used as a prerequisite to the more labourous "tabular method" by which to analyse the function's behaviour. The graph presented a curve with a vertical asymptote. The function's result from the left-hand side of the graph to the asymptote approaches negative infinity, and this same function's result from the right-hand side of the graph to the aysmptote approaches positive infinity. The **c**-coefficient is just a location parameter that shifts the vertical asymptot of the graph.

To demonstrate the function's end behaviour, I present the following table constructed by the labourous "tabular method", by which the function's input is presented in an extremely small absolute deviation from a value that woud cause the expression to be undefined.[2] I have set the **c**-parameter to 0, and the following table shows the behaviour of the function as it approaches the asymptote:

x	**f(x)**
-0.1	-21999999999.99999
-0.01	-2.1999999999999996e+19
-0.001	-2.1999999999999994e+28
-0.0001	-2.199999999999999e+37
-0.00001	-2.1999999999999983e+46
0.00001	2.1999999999999983e+46
0.0001	2.199999999999999e+37
0.001	2.1999999999999994e+28
0.01	2.1999999999999996e+19
0.1	21999999999.99999

From these data, I see that the resulting `f(x)` is lesser as the difference between "x" and "c" from the left-hand of the graph is smaller. Likewise, the resulting `f(x)` is greater as the difference between "x" and "c" from the right-hand of the graph is lesser. This has led me to draw the conclusion that:

$$\lim_{x \to c^+} = +\infty$$
$$\wedge \lim_{x \to c^-} = -\infty$$

Endnotes

1. Unfortunately, I cannot attach images or figures into this writeup, so I will have to use a verbalistic explaination and other methods to demonstrate end behaviour of the expression.
2. Specifically, if a value for "**x**" in the divisor of the expression results in 0, then the function will become undefined.

Ungraded **Grade:** 0/0.0

Total grade: 1.0×1/2 + 1.0×1/2 + 0.0×0/2 = 50% + 50% + 0%

Question4: *Score 1/1*

The Fourth Gate

You arrive at the fourth gate, feeling confident in your skills. "It is a good thing I had some practice before landing on this planet", you think. You look carefully at the gate and find the following information, translated by your computer:

Each of the limits below represent a piece of the passcode that you must enter to proceed. The parameters a and c represent positive real numbers and should be treated as such in your answer.

"Remember," says your computer, "to type "infinity" for ∞, "-infinity" for $-\infty$, and "NA" if the limit does not exist."

$$\lim_{x \to \infty} \frac{ax^{61} - 11x + 71}{cx^{61} - 71} =$$

Your response	Correct response
a/c	**a/c**

Auto graded **Grade:** 1/1.0

$$\lim_{x \to \infty} \frac{ax^{61} - 11x + 71}{cx^{58} - 71} =$$

Your response	Correct response
infinity	**infinity**

Auto graded **Grade:** 1/1.0

$$\lim_{x\to\infty} \frac{ax^{58}-11x+71}{cx^{61}-71} =$$

Your response	Correct response
0	**0**

Auto graded **Grade:** 1/1.0

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

Unlike with previous problems involving limits, I was able to use a more analytical solution to work out the resulting L-values for the given expressions. In particular, the limit is the function $f(x)$ as its input x values approaches infinity, or:

$$\lim_{x\to\infty} f(x)$$

In particular, three expressions are what define three variants of the $f(x)$, specifically:

$$f_1(x) = \lim_{x\to\infty} \frac{ax^{61} - 11x + 71}{cx^{61} - 71}\ (\forall a, c > 0)$$
$$f_2(x) = \lim_{x\to\infty} \frac{ax^{61} - 11x + 71}{cx^{58} - 71}\ (\forall a, c > 0)$$
$$f_3(x) = \lim_{x\to\infty} \frac{ax^{58} - 11x + 71}{cx^{61} - 71}\ (\forall a, c > 0)$$

Limit of $f_1(x)$

I used the following logic to evaluate the expression of the first function as $x \to \infty$:

$$\begin{cases} \lim_{x\to\infty} f_1(x) \\ = \lim_{x\to\infty} \frac{ax^{61}-11x+71}{cx^{61}-71} \\ = \lim_{x\to\infty} \frac{(ax^{61}-11x+71)\div x^{61}}{(cx^{61}-71)\div x^{61}} \\ = \lim_{x\to\infty} \frac{\frac{ax^{61}}{x^{61}}-\frac{11x}{x^{61}}+\frac{71}{x^{61}}}{\frac{cx^{61}}{x^{61}}-\frac{71}{x^{61}}} \\ = \lim_{x\to\infty} \frac{a-0+0}{c-0} = \frac{a}{c} \end{cases}$$

Limit of $f_2(x)$

I used the following logic to work out the limit as $x \to \infty$ for $f_2(x)$:

$$\begin{cases} \lim_{x\to\infty} f_2(x) \\ = \lim_{x\to\infty} \frac{ax^{61}-11x+71}{cx^{58}-71} \\ = \lim_{x\to\infty} \frac{(ax^{61}-11x+71)\div x^{58}}{(cx^{58}-71)\div x^{58}} \\ = \lim_{x\to\infty} \frac{\frac{ax^{61}}{x^{58}}-\frac{11x}{x^{58}}+\frac{71}{x^{58}}}{\frac{cx^{58}}{x^{58}}-\frac{71}{x^{58}}} \\ = \lim_{x\to\infty} \frac{ax^3-0+0}{c-0} = \frac{ax^3}{c} = \infty \end{cases}$$

Limit of $f_3(x)$

Finally, I used the following logic to evaluate the limit of $f_3(x)$ as $x \to \infty$:

$$\begin{cases} \lim_{x\to\infty} f_3(x) \\ = \lim_{x\to\infty} \frac{ax^{58}-11x+71}{cx^{61}-71} \\ = \lim_{x\to\infty} \frac{(ax^{58}-11x+71)\div x^{61}}{(cx^{61}-71)\div x^{61}} \\ = \lim_{x\to\infty} \frac{\frac{ax^{58}}{x^{61}}-\frac{11x}{x^{61}}+\frac{71}{x^{61}}}{\frac{cx^{61}}{x^{61}}-\frac{71}{x^{61}}} \\ = \lim_{x\to\infty} \frac{ax^{-3}-0+0}{c-0} = \lim_{x\to\infty} \frac{\frac{a}{x^3}}{c} \\ = \lim_{x\to\infty} \frac{ac}{x^3} = 0 \end{cases}$$

Ungraded **Grade:** 0/0.0

Total grade: 1.0×1/3 + 1.0×1/3 + 1.0×1/3 + 0.0×0/3 = 33% + 33% + 33% + 0%

Question5: *Score 0/1*

The Map

As you walk through Gate 4, you realize that Gate 5 is nowhere to be found. In front of you lies an expansive desert as far as you can see. You look at your map and see that there is an alert for this area. The warning states that this desert is almost entirely quicksand, with only one path safely through the desert. That path is defined by a piecewise function but requires that some parameters be determined before your computer can generate the plot. Thinking back on what you learned in calculus, you realize that your path will need to be continuous. You just need to tell your computer what the parameters below should be in order to create this continuous path. What parameters do you give your computer?

$$f(x) = \begin{cases} 6, & x \leq -2 \\ x + A, & -2 < x \leq -1 \\ Bx - 2, & -1 < x \leq 0 \\ 6x + C, & 0 < x \leq 1 \\ Dx + 6, & 1 < x \leq 2 \\ 2, & x > 2 \end{cases}$$

A	B	C	D

Your response	Correct response
	8
Auto graded	Grade: 0/1.0

Your response	Correct response
	-9
Auto graded	Grade: 0/1.0

Your response	Correct response
	-2
Auto graded	Grade: 0/1.0

Your response	Correct response
	-2
Auto graded	Grade: 0/1.0

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

Ungraded **Grade:** 0/0.0

Total grade: 0.0×1/4 + 0.0×1/4 + 0.0×1/4 + 0.0×1/4 + 0.0×0/4 = 0% + 0% + 0% + 0% + 0%

Question6: *Score 1/1*

The Fifth Gate

After traversing your continuous path, you finally arrive at Gate 5. Your computer reads the gate's directions to you. “The passcode for this gate is the number found by first entering $m'(-1)$ and then entering $n'(8)$ using the definitions below.” What values do you provide to your computer?

$$f(x) = \left(x^6 - 2x\right)$$

$$g(x) = \left(x^2 + 2x - 79\right)$$

$$h(x) = 2x - 15$$

$$m(x) = f(x) \cdot g(x)$$

$$n(x) = \frac{g(x)}{h(x)}$$

$$m'(-1) =$$

Your response	Correct response
640	**640**

Auto graded **Grade:** 1/1.0

$$n'(8) =$$

Your response	Correct response
16	**16**

Auto graded **Grade:** 1/1.0

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

The following simple functions are given:

$$f(x) = (x^6 - 2x)$$
$$g(x) = (x^2 + 2x - 79)$$
$$h(x) = (2x - 15)$$

The following more elaborate functions are also given:

$$m(x) = \frac{d}{dx}[f \times g]$$
$$n(x) = \frac{d}{dx}\left[\frac{g}{h}\right]$$

The problem is to work out the values of $m(-1)$ and $n(8)$. I have done so in the work

I. Work out the simpler derivatives

I shall "divide and conquer" such a problem by first working out the derivatives to the constituent expressions that make up the make up the larger `m(x)` and `n(x)` functions, which will be passed as "parameters" when evaluating their derivatives:

$$\frac{df}{dx} = 6x - 2$$
$$\frac{dg}{dx} = 2x + 2$$
$$\frac{dh}{dx} = 2$$

II. Derivatives of the `m(x)` and `n(x)` and their evaluations

The `m(x)` makes use of the product rule:

$$\frac{dm}{dx} = \frac{d}{dx}[f \times g] = \left[\frac{df}{dx}g + \frac{dg}{dx}f\right]$$

Specifically,

$$\frac{dm}{dx} = (6x - 2)(x^2 + 2x - 79) + (2x + 2)(x^6 - 2x)$$

When evaluating $\left.\frac{dm}{dx}\right|_{x=1}$, the arithmetic worked out to 640.

Next, regarding the derivative of n(x), it makes use of the quotient rule:

$$\frac{dn}{dx} = \frac{d}{dx}\left[\frac{g(x)}{h(x)}\right] = \frac{\frac{dg}{dx}h(x) - \frac{dh}{dx}g(x)}{h^2(x)}$$

Specifically,

$$\frac{dn}{dx} = \frac{(2x + 2)(2x - 15) - (2)(x^2 + 2x - 79)}{(2x - 15)^2}$$

When evaluating the $\left.\frac{dn}{dx}\right|_{x=8}$, the arithmetic worked out to 16.

Ungraded **Grade:** 0/0.0

Total grade: 1.0×1/2 + 1.0×1/2 + 0.0×0/2 = 50% + 50% + 0%

Question7: *Score 1/1*

The Sixth Gate

After walking through the fifth gate, your computer informs you that more than half of the gates have been crossed. "Only 4 left", you conclude. "Computer, what is this gate's riddle?"

Your computer responds:

For the function, $f(t) = \frac{1}{a}\left(6t^3 - 47t^2 + 141\right)^{44}$*, find the value of* a *that would make the slope of the tangent line* -176 *at* $t = 2$*.*

$a =$

Your response	Correct response
29	**29**

Auto graded **Grade:** 1/1.0

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

The following function is given:

$$f(t) = \frac{1}{a}\left(6t^3 - 47t^2 + 141\right)^{44}$$

The "riddle" is to work out a value for the a such that the $m_{\tan}$, or the slope of the tangent line of the function evaluated at an arbitrary point $t = 2$ results in $m_{\tan} = -176$. My solution to this problem requires two key steps: the first is to evaluate the derivative of `f(t)`, and the second is to work out a general formula to compute a given the resulting `f(t)` and target $m_{\tan}$ value.

I. Derivative of f(t)

This step involves applying a special case of the chain rule to power functions:

$$\begin{cases} \frac{1}{a}\left(6t^3 - 47t^2 + 141\right)^{44} \longleftarrow \text{Given} \\ = \frac{1}{a} \cdot \frac{d}{dt}\left[\left(6t^3 - 47t^2 + 141\right)^{44}\right] \\ = \frac{1}{a} \cdot \frac{d}{dt}\left[\left(6t^3 - 47t^2 + 141\right)^{44}\right] \\ = \frac{1}{a} \cdot \left[44 \cdot \left(6t^3 - 47t^2 + 141\right)^{44-1}\right] \cdot \frac{d}{dt}\left[6t^3 - 47 + 141\right] \\ = \frac{1}{a} \cdot (44 \cdot (6t^3 - 47t^2 + 141)^{43} \cdot (18t^2 - 94t) \end{cases}$$

II. General Solution to compute a given $m_{\tan}$, and a $\left.\frac{df}{dt}\right|_{t=t_i}$

For the sake of simplicity, I will define df as the key solution to the derivative:

$$df = 44 \cdot (6t^3 - 47t^2 + 141)^{43} \cdot (18t^2 - 94t)$$

The $m_{\tan}$ is worked out as

$$m_{\tan} = \left.\frac{df}{dt}\right|_{t=t_i}$$

Specifically, the slope of the tangent line for this model is:

$$m_{\tan} = \frac{1}{a}df$$

With a bit of algebra, I was able to solve in terms of a -

$$a = \frac{df}{m_{\tan}}$$

Setting $t = 2$ and $m_{\tan} = -176$, I have worked out the optimal value for "a" to be:

$$a = \frac{-5104}{-176} = 29$$

This can be double checked like so:

$$\begin{gathered} m_{\tan} = \frac{1}{a}df \;\; \text{(General Case)} \\ -176 = \frac{1}{29} \cdot df(2) \;\; \text{(Specific Case)} \\ \Rightarrow -176 = \frac{1}{29} \cdot -5104 \\ \Rightarrow -176 = -176 \end{gathered}$$

Ungraded **Grade:** 0/0.0

Total grade: 1.0×1/1 + 0.0×0/1 = 100% + 0%

Question8: *Score 1/1*

The Seventh Gate

As you approach the seventh gate, you notice that it is only a short distance to the area where your computer indicated the communication device was stored. You look past the gate hopefully, as your computer translates:

The passcode for this gate is found by determining the domain of the derivative of $f(x)$ and then evaluating the derivative at $x = -\frac{8}{22}$. Enter the domain in interval notation.

$f(x) = \sqrt{8 - 22x}$.

Domain of $f'(x)$:

Your response	Correct response
(-infinity, 4/11)	$\left(-\infty, \frac{4}{11}\right)$

Auto graded **Grade:** 1/1.0

$$f'\left(-\frac{8}{22}\right)$$

Your response	Correct response
-11/4	**-11/4**

Auto graded **Grade:** 1/1.0

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

Given the function:

$$f(x) = \sqrt{8-22x}$$

I attacked the problem by first expressing it in terms of a derivative:

$$\frac{d}{dx} = \frac{d}{dx}\left[\sqrt{8-22x}\right]$$

This can be solved by expressing the radical in terms of an exponent, and the applying a special case of the chain rule to the power law:

$$\frac{d}{dx}\left[\sqrt{8-22x}\right]$$
$$= \frac{d}{dx}\left[(8-22x)^{\frac{1}{2}}\right]$$
$$= \left[\frac{1}{2}(8-22x)^{\frac{1}{2}-\frac{2}{2}}\right] \cdot \frac{d}{dx}\left[8-22x\right]$$
$$= \frac{1}{2}(8-22x)^{-\frac{1}{2}} \cdot -22$$
$$= -\frac{22}{2}(8-22x)^{-\frac{1}{2}}$$
$$= -\frac{11}{1}\frac{1}{\sqrt{8-22x}} = -\frac{11}{\sqrt{8-22x}}$$

Next, I then proceeded to work out the domain of the $\frac{df}{dx}$, I did this by finding a solution to the inequality:

$$\sqrt{8-22x} > 0$$

The logic is that if the divisor of the resulting derivative is `0`, it will be undefined, and if `< 0`, then it would result in an imaginary number. The resulting value of $\frac{df}{dx}$ must be a real number, hence its divisor must result in a number > 0. The solution to this expression is:

$$\sqrt{8-22x} > 0$$
$$\Rightarrow (\sqrt{8-22x})^2 > (0)^2$$
$$\Rightarrow 8-22x > 0$$
$$\Rightarrow -22x > -8$$
$$\Rightarrow x < \frac{-8}{-22} \Rightarrow x < \frac{4}{11}$$

Therefore , the domain of the function is $\mathbb{D} := \left\{-\infty < x < \frac{4}{11}\right\}$

Finally, the derivative needs to be evaluated at $\frac{df}{dx}\Big|_{x=-\frac{8}{22}}$,

$$\frac{df}{dx}\Big|_{x=-\frac{8}{22}} = -\frac{11}{\sqrt{8-22\cdot\left(-\frac{8}{22}\right)}} = -\frac{11}{4}$$

Ungraded **Grade:** 0/0.0

Total grade: 1.0×1/2 + 1.0×1/2 + 0.0×0/2 = 50% + 50% + 0%

Question9: *Score 1/1*

The Eighth Gate

Arriving at the eighth and penultimate gate, you immediately get to work. "Computer, translate this gate for me." Your computer replies with the follow:

The function $f(t) = a\left(\sin\left(t^{10}\right)\right)^6$ models the position of an oscillating particle that has recently been discovered. Assume that a is a constant. What is the function that represents the velocity of this particle?

Note: If you want to write a power of a trigonometric function, remember that writing sin(x) to the fourth power as $\sin^4(x)$ is just a shorthand method. To be mathematically correct, you should write that as $(\sin(x))^4$ - which is what your computer expects and can translate.

Velocity Function:

Your response	Correct response
$60 \cdot a \cdot t^9 \cdot \sin^5\left(t^{10}\right) \cdot \cos\left(t^{10}\right)$	**60*a*t^9*cos(t^10)*sin(t^10)^5**

Auto graded **Grade:** 1/1.0

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

I went about setting up this problem as:

$$v(t) = \frac{d}{dt}\left[a \cdot \left(\sin\left(t^{10}\right)\right)^6\right]$$

This v(t) can be worked out with the chain rule:

$$v(t) = \frac{df}{dt} = \frac{d}{dt}\left[a \cdot \left(\sin(t^{10})\right)^6\right]$$

$$\Rightarrow \frac{d}{d[\cos(t)]}\left[a \cdot (fn_1)^6\right] \cdot \frac{d[\cos(t)]}{d[t^{10}]} \cdot \frac{d[t^{10}]}{dt}$$

$$= 6a(\sin^5(t^{10})) \cdot \cos\left(t^{10}\right) \cdot 10t^9$$

$$= 60at^9 \cdot \sin^5(t^{10}) \cdot \cos(t^{10})$$

where $fn_1 = \sin(t^{10})$

Ungraded **Grade:** 0/0.0

Total grade: 1.0×1/1 + 0.0×0/1 = 100% + 0%

Question10: *Score 1/1*

The Ninth Gate

You arrive at the final gate, which leads into a building-like structure. You hope that the communication device is past this last wall. You notice that there are tiles on this gate that can be moved and sorted. Your computer translates the gate with the following:

Order the tiles below from least to greatest. The smallest value must go to the left end and the largest value must go on the right end. Assume that a is a constant and that $a \geq 1$. Use the function definitions provided in organizing the tiles.

$f(x) = a^x \ln(x).$

$g(x) = \dfrac{\ln(x)}{\sqrt[7]{128x^3}}$

$h(x) = \dfrac{\ln(x)}{e^\pi}$

CLICK AND DRAG TO REARRANGE

Your response	Correct response

$h'(1)$	$g'(1)$	$f'(1)$	$h'(1)$	$g'(1)$

Auto graded **Grade:** 1/1.0

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

For `f(x)`, I applied the rules of the derivative calculus to work out its derivative:

$$(\forall a \geq 1) f(x) = a^x \cdot \ln(x)$$
$$\Rightarrow \frac{df}{dx} = \frac{d}{dx}\left[a^x \ln(x)\right] \Rightarrow \frac{df}{dx} = \frac{d[a^x]}{dx} \cdot \ln(x) + \frac{d[\ln(x)]}{dx} \cdot a^x$$
$$\Rightarrow \frac{df}{dx} = \left\{a \geq 1 : a^x \ln(a) \ln(x) + \frac{a^x}{x}\right\}$$

For `g(x)` and `h(x)`, I applied a numerical solution of the derivative which involves setting the Δx constituent to an arbitrarily small number:

$$\frac{df}{dx} = \operatorname{argmin}_{|\Delta x - 0|} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Applying this numeric solution to `g(x)` and `h(x)`, it was yielded:

$$\frac{dg}{dx} = \frac{g(1 + \Delta x) - g(1)}{\Delta x} = \frac{1}{2}$$
$$\frac{dh}{dx} = \frac{g(1 + \Delta x) - g(x)}{\Delta x} = \frac{\ln(2)}{e^\pi} \approx 0.29953$$

If the lesser value is to be placed first, and the greater value to the right-most, then the proper order is:

$$h'(1) < g'(1) < f'(1)$$

Ungraded **Grade:** 0/0.0

Total grade: 1.0×1/1 + 0.0×0/1 = 100% + 0%

Question11: *Score 0/1*

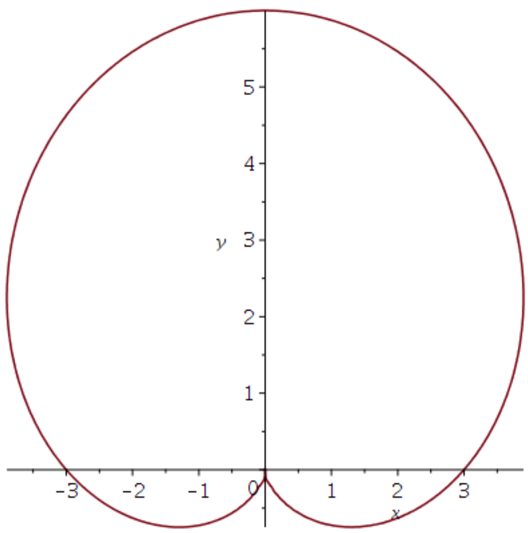
Aiming the Communication Device

You reach the communication device and are alerted by your computer that the communication device is surrounded by a force field that can be described by the equation, $x^2 + y^2 = \left(\frac{1}{3}x^2 + \frac{1}{3}y^2 - y\right)^2$, with the origin being where you located this device. The inhabitants of this planet did not want unwelcomed guests, so they made sure to guard the use of this device. Your computer helpfully generates a plot of this equation and tells you that it is a cardioid (whatever that means). The communication device can be taken to the very edge of the force field, but not beyond it. In fact, the only way to use the device is to align it tangent to the force field. Your computer also recovers some files that give points along the force field where the communication device can be used to communicate with different planets. Communication with your home planet can be achieved from the point on the force field, $(3, 0)$. If you provide your computer with the equation of the line that satisfies these conditions, you will be able to send a message to your home world. What equation do you provide your computer?

Equation of the Line:

Your response	Correct response
No answer	**y= x-3**

Auto graded **Grade:** 0/1.0



Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

Ungraded **Grade:** 0/0.0

Total grade: 0.0×1/1 + 0.0×0/1 = 0% + 0%

Question12: *Score 1/1*

Connecting the Communication Device

With the communication device pointed toward your home planet, you realize, to your dismay, that the cords have been disconnected from the device. "Computer, how do I plug in these cables", you ask. Your computer responds that the directions (translated into mathematics) suggest that you order the cables from smallest to largest based on the below rules. Port 1 gets the smallest value and Port 3 gets the largest value, where it is assumed that c is a constant such that $c > 1$.

$f(x) = \ln(2x)$

$g(x) = 4\sin(3x)$

Red Cord: $f'(c)$

Green Cord: $g^{(8)}(3\pi)$

Blue Cord: $f''(c)$

CLICK AND DRAG TO REARRANGE

Port 1	Port 2	Port 3

Your response			Correct response	
Blue Cord: $f''(c)$	Green Cord: $g^{(8)}(3\pi)$	Red Cord: $f'(c)$	Blue Cord: $f''(c)$	Green Cord: $g^{(8)}(3\pi)$

Auto graded **Grade:** 1/1.0

Explain, in your own words and with your own work, how you arrived at this result. Be sure to explain using calculus concepts to best support the work of the game design team.

For `f(x)`, I used the following approach to work out its derivative:

$$\frac{d}{dx}\Big[\ln(2x)\Big] = \frac{1}{x}$$
$$\frac{d^2 f}{dx} = \frac{d}{dx}\Big[-\frac{1}{x^2}\Big]$$

Now, the formula that I used to work out the

$$\left.\frac{d^8 g}{dx^8}\right|_{x=3\pi} 4\sin(3x)$$
$$= \begin{cases} wa^k \cdot f(x) & \text{if } k \bmod 4 = 0 \\ wa^k \cdot \dfrac{d^{k \bmod 4} f}{dx^{k \bmod 4}} & \text{otherwise} \end{cases}$$
$$\text{specifically } = 4 \cdot 3^8 \cdot \sin(3x)$$
$$\Rightarrow \left.\frac{d^8 f}{dx^8}\right|_{x=3\pi} = 0$$

Assuming that $c > 1$ for the input, the first derivative of `f(x)` will always be `> 0`, second will always be `< 0`, and the eighth derivative of `g(x)` will always work out to `0`. So, if these "ports" are to be organised in a manner where the first one gets the lesser value, the thrid one gets the greater value, and the second one gets the mediocre value, the proper order would therefore be:

$$\left.\frac{d^2 f}{dx^2}\right|_{x_C > 1} < \left.\frac{d^8 g}{dx^8}\right|_{x=3\pi} < \left.\frac{df}{dx}\right|_{x_C > 1}$$

$$\text{Or, in the context of the given problem ...}$$

$$\text{Blue Chord} < \text{Green Chord} < \text{Red Chord}$$

Ungraded **Grade:** 0/0.0

Total grade: 1.0×1/1 + 0.0×0/1 = 100% + 0%